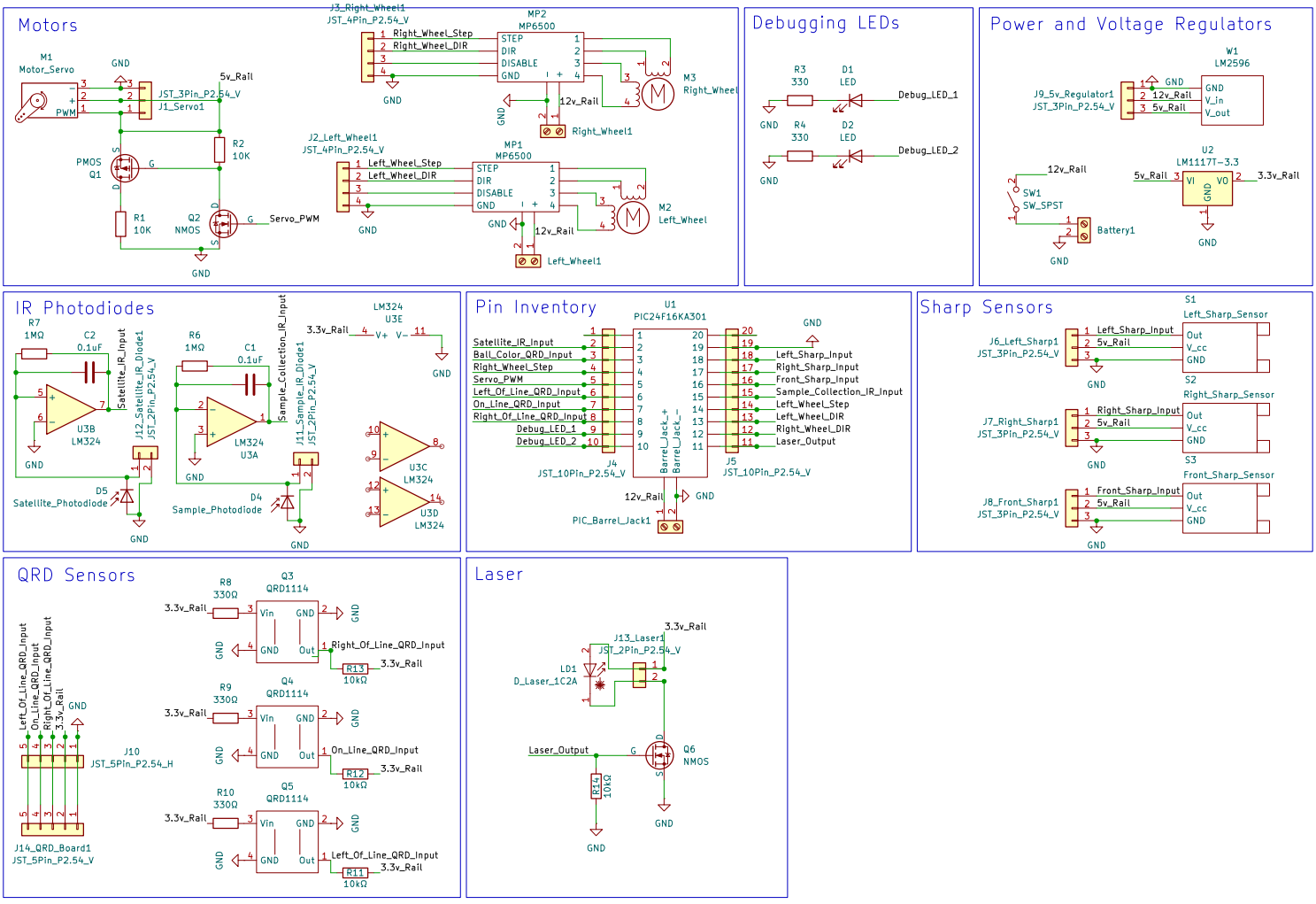

Motors
M1
Motor_Servo
PWM
GND
5v_Rail
J1_Servo1
JST_3Pin_P2.54_V
PMOS
Q1
R2
10K
R1
10K
Q2
NMOS
Servo_PWM
J3_Right_Wheel1
JST_4Pin_P2.54_V
Right_Wheel_Step
Right_Wheel_DIR
MP2
MP6500
STEP
DIR
DISABLE
GND
12v_Rail
M3
Right_Wheel
Right_Wheel1
J2_Left_Wheel1
JST_4Pin_P2.54_V
Left_Wheel_Step
Left_Wheel_DIR
MP1
MP6500
M2
Left_Wheel
Left_Wheel1
Debugging LEDs
R3
330
D1
LED
Debug_LED_1
R4
330
D2
LED
Debug_LED_2
Power and Voltage Regulators
W1
LM2596
J9_5v_Regulator1
JST_3Pin_P2.54_V
GND
12v_Rail
5v_Rail
V_in
V_out
U2
LM1117T-3.3
VI
VO
GND
3.3v_Rail
SW1
SW_SPST
Battery1
IR Photodiodes
R7
1MΩ
C2
0.1uF
Satellite_IR_Input
J12_Satellite_IR_Diode1
JST_2Pin_P2.54_V
U3B
LM324
D5
Satellite_Photodiode
R6
1MΩ
C1
0.1uF
Sample_Collection_IR_Input
J11_Sample_IR_Diode1
JST_2Pin_P2.54_V
U3A
LM324
D4
Sample_Photodiode
LM324
U3E
3.3v_Rail
U3C
LM324
U3D
LM324
Pin Inventory
U1
PIC24F16KA301
Satellite_IR_Input
Ball_Color_QRD_Input
Right_Wheel_Step
Servo_PWM
Left_Of_Line_QRD_Input
On_Line_QRD_Input
Right_Of_Line_QRD_Input
Debug_LED_1
Debug_LED_2
Barrel_Jack_+
Barrel_Jack_-
J4
JST_10Pin_P2.54_V
J5
JST_10Pin_P2.54_V
Left_Sharp_Input
Right_Sharp_Input
Front_Sharp_Input
Sample_Collection_IR_Input
Left_Wheel_Step
Left_Wheel_DIR
Right_Wheel_DIR
Laser_Output
12v_Rail
PIC_Barrel_Jack1
Sharp Sensors
S1
Left_Sharp_Sensor
J6_Left_Sharp1
JST_3Pin_P2.54_V
Left_Sharp_Input
5v_Rail
GND
S2
Right_Sharp_Sensor
J7_Right_Sharp1
JST_3Pin_P2.54_V
Right_Sharp_Input
S3
Front_Sharp_Sensor
J8_Front_Sharp1
JST_3Pin_P2.54_V
Front_Sharp_Input
QRD Sensors
R8
330Ω
Q3
QRD1114
3.3v_Rail
Vin
GND
Out
Right_Of_Line_QRD_Input
R13
10kΩ
R9
330Ω
Q4
QRD1114
On_Line_QRD_Input
R12
10kΩ
R10
330Ω
Q5
QRD1114
Left_Of_Line_QRD_Input
R11
10kΩ
J10
JST_5Pin_P2.54_H
J14_QRD_Board1
JST_5Pin_P2.54_V
Laser
J13_Laser1
JST_2Pin_P2.54_V
LD1
D_Laser_1C2A
Laser_Output
Q6
NMOS
R14
10kΩ